ENLARGING TWO BOUNDED SUBCOMPLEXES TO A COMMON PAIR THAT KILLS A DERIVED-TENSOR KERNEL AND CATCHES AN IMAGE, INSIDE A CARDINAL BOUND

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Editorial note. The proof contains two occurrences of the phrase Details omitted, at source lines 29953 and 29963, concerning the size estimate for the smallest subcomplexes generated by finitely many elements and the existence of a free resolution of B of cardinality at most kappa. Both are elementary supporting steps, neither is a whole block of the main line, and the main argument from the choice of kappa to the cardinal bound of property (3) is fully present in the counted span; the phrase is preserved verbatim as written by the author.

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Editorial note. Property (2) as printed by the author ends in H^2(N_2) where the surrounding text suggests H^i; the printed text is kept verbatim, since the lane copies the author's statement and neither corrects nor comments on its mathematics.

Allowed standard prerequisites (editorial).

- Arithmetic of cardinals: the maximum of a finite family of cardinals bounds the cardinality of a union or of a finite product of sets of those cardinalities. invoked at proof lines 29924-29932, where kappa = max(|A|, |B|, aleph_0) is fixed and the statement is announced to be used without further mention. Supplies the size estimate for the smallest subcomplexes generated by finitely many elements, for the free resolution of B, and finally for the

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index set of the two sums that define M_2 and N_2, which is what makes property (3) true.

- Every module admits a resolution by free modules of bounded cardinality, and the derived tensor product is computed by the total complex of the tensor product with such a resolution. recorded at proof lines 29962-29965 as a resolution ... -> F^{-1} -> F^0 -> B by free A-modules F_i with |F_i| at most kappa, with the cohomology of M_1 tensor B computed by Tot(M_1 tensor F). Turns the abstract derived tensor product into an explicit complex of cardinality at most max(kappa, |M_1|), which is the estimate used for the cohomology of the small subcomplexes.
- Derived tensor product commutes with filtered colimits, so a cohomology class that dies after tensoring with B already dies after tensoring with a small enough subcomplex. used at proof lines 29955-29959, where M and N are exhibited as filtered colimits of the small subcomplexes, and at proof lines 29975-29978. Produces the finite collections E_xi and F_xi for a kernel class xi, and the analogous collections E_eta and F_eta for an image class eta, inside the small colimit system.
- The hypothesis that a : M -> N induces an isomorphism M tensor^L_A B -> N in D(B) is an isomorphism on cohomology, and it factors through the map to N_1. used at proof lines 29972-29974 for the kernel class xi and at proof lines 29981-29983 for the image class eta. Identifies the target of the vanishing condition for xi and the source of the preimage for eta with objects living in the N_1 and M_1 systems, which is what lets the enlargement be driven by finitely many elements.

Lemma 0.2. *Let $A \to B$ be a ring map. There exists a cardinal $\kappa = \kappa(A \to B)$ with the following property: Let $M^\bullet$, resp. $N^\bullet$ be a complex of A-modules, resp. B-modules. Let $a : M^\bullet \to N^\bullet$ be a map of complexes of A-modules which induces an isomorphism $M^\bullet \otimes^{\mathbf{L}}_A B \to N^\bullet$ in $D(B)$. Let $M_1^\bullet \subset M^\bullet$, resp. $N_1^\bullet \subset N^\bullet$ be a subcomplex of A-modules, resp. B-modules such that $a(M_1^\bullet) \subset N_1^\bullet$. Then there exist subcomplexes*

$$M_1^\bullet \subset M_2^\bullet \subset M^\bullet \quad \text{and} \quad N_1^\bullet \subset N_2^\bullet \subset N^\bullet$$

such that $a(M_2^\bullet) \subset N_2^\bullet$ with the following properties:

1. *$\mathrm{Ker}(H^i(M_1^\bullet \otimes^{\mathbf{L}}_A B) \to H^i(N_1^\bullet))$ maps to zero in $H^i(M_2^\bullet \otimes^{\mathbf{L}}_A B)$,*
2. *$\mathrm{Im}(H^i(N_1^\bullet) \to H^i(N_2^\bullet))$ is contained in $\mathrm{Im}(H^i(M_2^\bullet \otimes^{\mathbf{L}}_A B) \to H^2(N_2^\bullet))$,*
3. *$|\bigcup M_2^i \cup \bigcup N_2^i| \leq \max(\kappa, |\bigcup M_1^i \cup \bigcup N_1^i|)$.*

Proof. Let $\kappa = \max(|A|, |B|, \aleph_0)$. Set $|M^\bullet| = |\bigcup M^i|$ and similarly for other complexes. With this notation we have

$$\max(\kappa, |\bigcup M_1^i \cup \bigcup N_1^i|) = \max(\kappa, |M_1^\bullet|, |M_2^\bullet|)$$

for the quantity used in the statement of the lemma. We are going to use this and other observations coming from arithmetic of cardinals without further mention.

First, let us show that there are plenty of "small" subcomplexes. For every pair of collections $E = \{E^i\}$ and $F = \{F^i\}$ of finite subsets $E^i \subset M^i$, $i \in \mathbf{Z}$ and $F^i \subset N^i$, $i \in \mathbf{Z}$ we can let

$$M_1^\bullet \subset M_1(E,F)^\bullet \subset M^\bullet \quad \text{and} \quad N_1^\bullet \subset N_1(E,F)^\bullet \subset N^\bullet$$

be the smallest subcomplexes of A and B-modules such that $a(M_1(E,F)^\bullet) \subset N_1(E,F)^\bullet$ and such that $E^i \subset M_1(E,F)^i$ and $F^i \subset M_2(E,F)^i$. Then it is easy to see that

$$|M_1(E,F)^\bullet| \leq \max(\kappa, |M_1^\bullet|) \quad \text{and} \quad |M_2(E,F)^\bullet| \leq \max(\kappa, |M_2^\bullet|)$$

Details omitted. It is clear that we have

$$M^\bullet = \mathrm{colim}_{(E,F)} M_1(E,F)^\bullet \quad \text{and} \quad N^\bullet = \mathrm{colim}_{(E,F)} N_1(E,F)^\bullet$$

and the colimits are (termwise) filtered colimits.

There exists a resolution $\ldots \to F^{-1} \to F^0 \to B$ by free A-modules F_i with $|F_i| \leq \kappa$ (details omitted). The cohomology modules of $M_1^\bullet \otimes_A^{\mathbf{L}} B$ are computed by $\mathrm{Tot}(M_1^\bullet \otimes_A F^\bullet)$. It follows that $|H^i(M_1^\bullet \otimes_A^{\mathbf{L}} B)| \leq \max(\kappa, |M_1^\bullet|)$.

Let $i \in \mathbf{Z}$ and let $\xi \in H^i(M_1^\bullet \otimes_A^{\mathbf{L}} B)$ be an element which maps to zero in $H^i(N_1^\bullet)$. Then ξ maps to zero in $H^i(N^\bullet)$ and hence ξ maps to zero in $H^i(M^\bullet \otimes_A^{\mathbf{L}} B)$. Since derived tensor product commutes with filtered colimits, we can find finite collections E_ξ and F_ξ as above such that ξ maps to zero in $H^i(M_1(E_\xi, F_\xi)^\bullet \otimes_A^{\mathbf{L}} B)$.

Let $i \in \mathbf{Z}$ and let $\eta \in H^i(N_1^\bullet)$. Then the image of η in $H^i(N^\bullet)$ is in the image of $H^i(M^\bullet \otimes_A^{\mathbf{L}} B) \to H^i(N^\bullet)$. Hence as before, we can find finite collections E_η and F_η as above such that η maps to an element of $H^i(N_1(E_\eta, F_\eta)$ which is in the image of the map $H^i(M_1(E_\eta, F_\eta)^\bullet \otimes_A^{\mathbf{L}} B) \to H^i(N_1(E_\eta, F_\eta)$.

Now we simply define

$$M_2^\bullet = \sum\nolimits_\xi M_1(E_\xi, F_\xi)^\bullet + \sum\nolimits_\eta M_1(E_\eta, F_\eta)^\bullet$$

where the sum is over ξ and η as in the previous two paragraphs and the sum is taken inside $M^\bullet$. Similarly we set

$$N_2^\bullet = \sum\nolimits_\xi N_1(E_\xi, F_\xi)^\bullet + \sum\nolimits_\eta N_1(E_\eta, F_\eta)^\bullet$$

where the sum is taken inside $N^\bullet$. By construction we will have properties (1) and (2) with these choices. The bound (3) also follows as the set of ξ and η has cardinality at most $\max(\kappa, |M_1^\bullet|, |N_1^\bullet|)$. □